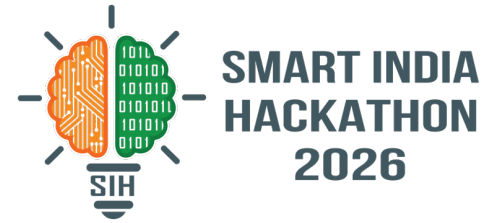
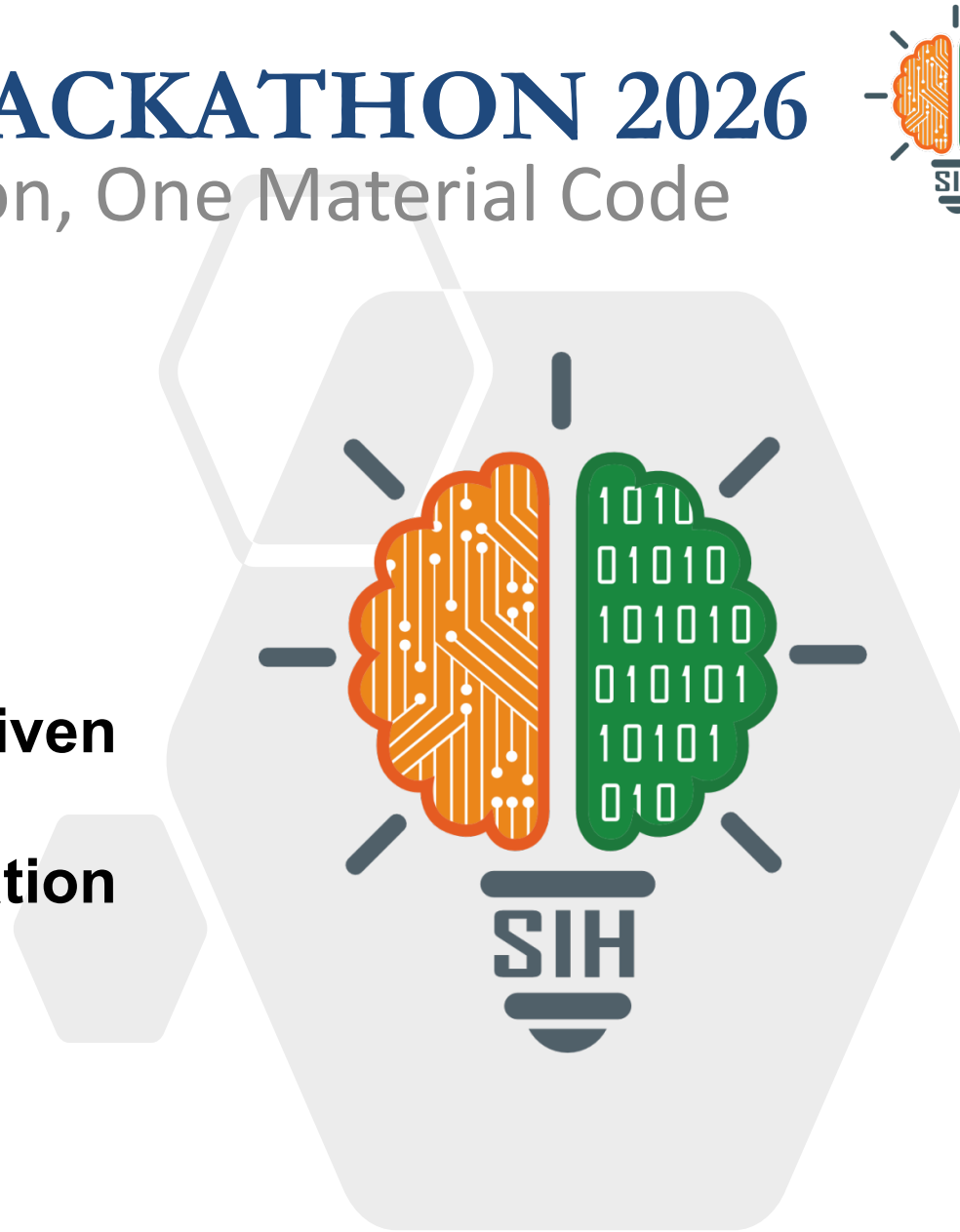


SMART INDIA HACKATHON 2026

UnifyMat — One Nation, One Material Code



- **Problem Statement ID – SIH26099**
- **Problem Statement Title- AI-Driven Standardization and Harmonization of Material Codes Across CPSEs**
- **Theme- Smart Automation**
- **PS Category- Software**



UnifyMat

One Nation, One Material Code

❖ Proposed Solution

- One National Material Code (NMC) per real material, all CPSEs
- Same bolt today: 5 codes, 5 prices (₹16–₹21) — merged safely
- AI proposes · hard facts veto · humans approve

Proven on 403 records · 5 CPSEs: precision 1.000 · 0 wrong merges · ~10 s on CPU

5 CPSE material masters → one national code

CPCL

IOCL

NTPC

SAIL

GAIL



ONE NATIONAL MATERIAL CODE
NMC-FAST-BOLT-9BF0ABAC

TECHNICAL APPROACH

- Tech: Python, Streamlit, MiniLM embeddings, TF-IDF fallback, RapidFuzz
- Flow: normalize → 35 attrs → match & veto → deterministic NMC + audit



Your
Team
Name

FEASIBILITY AND VIABILITY



- Feasible: any ERP CSV in; ~10 s per run
- Risk: wrong merges — veto + officer gate
- Proved on ground-truthed synthetic data

403 → 264

national codes issued

0

trap violations (8.8 vs 10.9)

7,951

unsafe pairs auto-rejected

IMPACT AND BENEFITS

- 300+ CPSEs on one national material language
- Safety: 8.8 vs 10.9 mix-up now impossible

**Economic: 12.8% avg price spread exposed — pump ₹26,674 vs ₹54,000 (51%);
aggregated demand → national rate contracts**

1.000

auto-merge precision

0

wrong merges

89.7%

recall with approvals

Every merge is officer-approved — named, timestamped, fully auditable.

- Problem Statement SIH26099 — MoP&NG / CPCL
 - Reimers & Gurevych (2019) — Sentence-BERT sentence embeddings · [arXiv:1908.10084](https://arxiv.org/abs/1908.10084)
 - all-MiniLM-L6-v2 model · huggingface.co/sentence-transformers/all-MiniLM-L6-v2
 - Papadakis et al. (2020) — Blocking & Filtering for Entity Resolution · ACM CSUR
 - ISO 898-1 — fastener property classes 8.8 / 10.9 (the grade trap in our data)
 - Government e-Marketplace (GeM) · gem.gov.in — national public-procurement context
 - Working prototype, benchmark + verified outputs · github.com/Varunsai1930/SIH